

Safety Management System Implementation Plan



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Portland International Airport Safety Management System Implementation Plan

The Port of Portland (Port) owns and operates the Portland International Airport (PDX or Airport). PDX is classified as a medium hub airport; the Airport serves the Portland metropolitan area and surrounding communities of Oregon and southwest Washington. PDX is a joint civil-military airport and the largest airport in Oregon state. The Oregon Air National Guard maintains a base on the southeast portion of the airport and hosts the 142d Fighter Wing which operates the F-15 Eagle. PDX operates three runways 10R/28L (11,000 ft x 150 ft), 10L/28R (9,825 ft x 150 ft), and 03/21 (6,000 ft x 150 ft.).

SMS Development and Deployment Strategy

I. SMS Development

From February 2017 through March 2018 the Port contracted with a consultant to help conduct a Safety Management System (SMS) review and gap analysis. The gap analysis was performed to determine the level of effort necessary to implement a compliant SMS that would align with the future Federal Aviation Administration (FAA) Part 139 SMS rulemaking. The gap analysis also took into consideration the potential integration of the Port's enterprise-wide occupational safety initiative into a single safety system that would combine the Safety and Airside Operations team efforts into a joint Safety/SMS endeavor. Findings and recommendations founded on industry expertise and best practices were presented in a Gap Report. In addition to a Gap Report, a draft SMS Manual and a draft Safety Plan were developed.

The SMS initiative was postponed after the 2018 rulemaking was delayed. In 2023, Port staff assessed the previous gap analysis against final regulatory requirements to update the gap analysis and create the Implementation Plan (IP). The SMS IP was developed in compliance with the requirements set forth in FAR Part 139.403(B). The most significant change in the current PDX Safety/SMS implementation approach is a separation of occupational and operational safety functions. Much of the original gap information will be leveraged to develop the SMS Program and SMS Manual.

- **How will the airport develop its SMS Manual? Will it procure consultant assistance or develop it in-house?**

The Port has an existing draft SMS Manual and will update it based on the Advisory Circular (AC) 150_5200_37A and Part 139 Subpart E requirements. The SMS Manual was originally developed in 2018 with the assistance of an SMS consultant who is also supporting the Port in the current update to the gap analysis and IP. The SMS Manual includes all SMS components and elements; however, new requirements from the AC and Subpart E need to be incorporated and updated within the current PDX operations. Development of the SMS Manual will be led by dedicated, internal staff. Outside consultant support will be procured as needed.

- **Will the airport conduct a formal gap analysis? If so, what is the timeline for completion?**

The current SMS gap analysis effort was completed between June 2023 and March 2024 using a detailed SMS questionnaire to compare prior gap information, existing operations, and status. The previous gap analysis was completed in 2018. The updated gap assessment is applying the new AC 150_5200_37A and Subpart E requirements to ensure compliance with SMS. A Development Plan will be created from the gap analysis effort to outline all tasks, actions, and activities required to complete and deploy the SMS. The gap analysis was used to inform the development of the SMS Implementation Plan and will be used during development of the SMS program.

Are there existing programs, policies, or practices that the airport plans to use as a foundation for the SMS elements?

Existing tools, policies, and procedures will be modified to meet the requirements of SMS. Where appropriate, components from the existing (Occupational) Safety Program will be used as a foundation for the SMS. Additionally, the two safety programs (SMS and Occupational Safety) will be differentiated and managed as two separate programs with collaboration as appropriate.

The Port plans to use existing software, VEOCI, to integrate the SMS into the existing reporting, tracking, and work order process. The Port will also incorporate SMS into existing staff and tenant meetings and will expand existing communication methods and formats for staff and tenant outreach.

II. SMS Deployment Strategy

- **Does the airport plan to use a phasing strategy? If so, what will be phased (i.e., phasing the SMS into the movement area first and then into the non-movement area or phasing the components and elements of SMS)?**

The Port does not intend to use a phasing strategy. The Port plans to develop the SMS uniquely for airside operations (movement and non-movement areas) and will develop the SMS components in a sequential manner building upon the information and documentation completed during the current gap analysis and IP effort. The Port will integrate the SMS components and requirements with the future VEOCI SMS software module to ensure the software supports the Airport's planned processes and procedures as well as FAA requirements.

- **Will the airport include its landside operations in the SMS? If so, how will landside requirements be kept separate from airside requirements?**

As part of the initial launch, the SMS will focus on airside operations only. This will include both movement and non-movement areas. This approach is intended to ensure compliance with the requirements set forth in FAR Part 139 Subpart E.

Hazard reporting and analysis is currently conducted under the existing PDX Occupational Safety program. With the launch of SMS, safety risk assessments, specific to airside operations, will be managed under the SMS. Other hazard and safety risk assessments will be addressed through the existing PDX Safety

Program. Incoming hazard reports will be triaged by the SMS Team and directed to the appropriate program (Safety or SMS). The software system enables reporting by location to ensure only airside records are delivered to the FAA.

III. Procurement

- **Does the airport plan to acquire any new systems or technology that will require procurement? If so, how long does the airport anticipate that procurement taking?**

The Port plans to use the VEOCI platform to support the SMS. PDX is an existing user of VEOCI in several departments, including Airside Operations. The VEOCI SMS module will be procured and integrated into the existing VEOCI functionality. Procurement of the VEOCI SMS module is being evaluated by the PDX Information Technology Team and Management to ensure the module cost, features, and configuration will support SMS functions and align with the Implementation Plan schedule.

Schedule for SMS Development and Deployment:

I. Major Milestones

The Port developed the following milestones and target dates. The Airport assessed reasonable dates for development and rollout of the SMS based on staffing and system availability, and included existing SMS resources such as the SMS Manual as part of the process review.

Milestone	Target Date
<i>Develop Implementation Plan</i>	April 15, 2024
The Airport created the IP using a detailed set of questions that covered all aspects of SMS planning, scheduling, development, and deployment. Questions were founded on best practices from previous Airport SMS Pilot Studies, Part 139 Subpart E, the FAA Advisory Circular 150_5200_37A, and the Final FAA Rulemaking. Responses took into consideration existing Airport functions, processes, and procedures to support the SMS. Port management reviewed and approved the IP.	Completed
<i>Conduct Gap Analysis</i>	March, 2024
Meetings with relevant departments were conducted in September and October of 2023 and collected additional information and documentation to prepare the gap analysis and IP. Staff reviewed the findings and based on this information, drafted the IP response, and developed the SMS milestones, targets, and program timeline.	Completed
<i>Finalize SMS Manual</i>	12 Months from approval of Implementation Plan
The Port plans to begin updating and revising the existing draft upon FAA's approval of the Implementation Plan. The gap analysis resulted in a checklist of missing SMS Manual components and elements to be incorporated into the revised Manual. Key updates include revisions to the safety policy statement, organizational structure and management/staff responsibilities, risk matrix, hazard reporting, promotional items, training, and evaluations/audits.	

<i>Provide SMS Training</i>	Dev Team: Ongoing Staff: March, 2026 Tenants: June, 2026
The Port's strategy for training includes hiring an SMS Manager in mid-2024 to manage the SMS and share expertise with Port staff and management. The SMS development team and Port management will participate in briefings and gain expertise as the SMS is implemented. The SMS Manager will develop a training and orientation plan to identify skills gaps in staff and to create an outline of required courses and workshops to deliver to Port staff and management. Relevant staff will receive training to support the internal SMS rollout in January of 2026 and all staff and tenants will receive orientation training concurrent to the SMS rollout in June 2026.	
<i>Deploy Hazard Reporting System</i>	Rollout June, 2026
Initially, hazard reporting will be implemented internally to enable testing of the system and to ensure processes are in alignment with the SMS Manual. The Port plans to purchase the VEOCI SMS module in the Spring of 2024. The SMS module includes a customizable web-based hazard reporting form. The Port plans to test the SMS module with staff starting in March of 2026 then deploy the hazard reporting system to all tenants by June 2026 with tenant orientation and promotional strategies deployed simultaneously.	
<i>Full Implementation of SMS</i>	36 months following FAA approval of SMS IP.
The planned target date for full implementation is May 2027.	
<i>Conduct First Evaluation of SMS Implementation</i>	July, 2027
The Port plans to complete an evaluation of the SMS in anticipation of the first Part 139 Certification Inspection which will include the SMS. Port staff will log findings as part of the evaluation and make corrections prior to the Certification Inspection.	

II. Challenges

The Port does not anticipate challenges to the SMS development and implementation. With management support and active engagement, the Port plans to leverage previous SMS documentation, fund the SMS, engage consultant services to assist with the development effort, and assign dedicated staff to deploy the SMS, maintain the project schedule, and monitor compliance.



SMS Implementation Plan Checklist:

<i>Item and Task</i>	<i>Description of Any Existing Systems / Processes Being Used to Meet the Task</i>	<i>Status of Implementation</i>	<i>Target for Completion</i>
1. Documentation			
1.1 SMS Manual	During the 2017 to 2018 gap analysis and Implementation effort, a draft SMS Manual was created. The SMS Manual will be updated to reflect the current requirements outlined in FAR Part 139 Subpart E and AC 150_5200_37A.	The Draft 2018 SMS Manual will be updated as part of the SMS Manual deliverable to FAA.	June, 2025
2. Safety Policy			
2.1 Identify the Accountable Executive	The Port has identified a suite of SMS management positions at various levels to ensure communication and escalation throughout the Airport's airside operations. Roles and titles are listed below: ▪ The Accountable Executive (AE) has been identified. The AE will be the Port's Chief Aviation Officer. ▪ The Responsible Executive (RE) is an additional leadership position to oversee the day-to-day SMS within the Airport's operations. The RE will be the Director of Airport Operations.	Completed	Completed

<i>Item and Task</i>	<i>Description of Any Existing Systems / Processes Being Used to Meet the Task</i>	<i>Status of Implementation</i>	<i>Target for Completion</i>
	The Port has also identified a third management position, the Responsible Manager (RM) will oversee staff and compliance for Part 139. The RM will be the Senior Manager PDX Airside / GA Airports.	Completed	Completed
2.2 Develop and distribute a Safety Policy Statement	The Port has an existing PDX Safety Policy focused on occupational safety. The Port is in the process of developing a separate SMS Safety Policy Statement that will align with SMS operational safety. The Policy statement will be endorsed and signed by the SMS Management positions (AE, RE, and RM). The SMS Safety Policy statement will be hosted on the PDX website. It will also be cited in SMS training and presented in stakeholder meetings.	A new SMS (operational) safety policy statement will be authored, reviewed, and approved by management as part of the updated SMS Manual development effort.	June, 2025
2.3 Make the Safety Policy Statement (SPS) available to all employees and tenants	The Safety Policy Statement will be accessible to all employees and tenants through the PDX website. A new set of pages will be developed to host SMS information including the Policy. The Policy statement will be included in all briefings, training, and orientation materials. Distribution will be tracked in the Safety Communications Log. (See 5.7)	To align with staff rollout, the safety policy statement will be promoted at the same time.	Staff SMS Rollout January, 2026

<i>Item and Task</i>	<i>Description of Any Existing Systems / Processes Being Used to Meet the Task</i>	<i>Status of Implementation</i>	<i>Target for Completion</i>
2.4 Identify organizational structure responsible for airport safety issues	The organization structure to support the SMS is under consideration. Initially, this will include an SMS Manager who will report to the Senior Manager PDX Airside / GA Airports (Responsible Manager). The AE will be the Chief Aviation Officer and the RE will be the Director of Airport Operations. The Port will create a new internal SMS Steering Committee that will include Port leadership, including representatives from key PDX departments. For tenant engagement, the existing PDX Tenant Safety Advisory Group (TSAG) will be expanded to include SMS as a new responsibility. An SMS sub-committee will be developed to enable Port-tenant discussions regarding airfield safety concerns, hazard reports, mitigation strategies, etc.	The safety organizational is largely defined at the management level; however, when the new SMS Manager is hired a final organizational structure will be completed that includes collateral duties and collaborations from other departments as well as all SMS-related committees.	December, 2024

2.5 Communicate safety organizational structure to airport employees and managers	Communication of the organizational structure will be incorporated into all levels of SMS training including both internal (Port employees) and tenant orientations. As the SMS is being developed, staff will be informed of their new responsibilities through meetings, briefings, and standing committees. The safety organizational structure will also be described in the SMS policy documents available to all airport employees and tenants and will be included in the SMS Manual.	The safety organization will be communicated after the final organizational structure is completed (see 2.4).	December, 2024
2.6 Define airport management, including various levels throughout the organization, responsibilities, and accountabilities for safety issues	▪ See 2.1 for AE, RE, and RM positions, titles, and responsibilities. ▪ SMS Manager – Overall management and responsibility for the SMS implementation and operation including strategies, documentation, and compliance. ▪ Airport Duty Managers (ADMs) and Airport Operations Specialists (AOSs) – will support the SMS as part of their day-to-day responsibilities. ▪ Maintenance Managers will support SMS mitigations and safety assessments as experts and will also provide insights on safety hazards. ▪ Security, Environmental, Properties, Legal, Risk, and Occupational Safety as well as other departments will support the SMS as secondary roles providing expertise and guidance	See 2.3 and 2.4 the final organizational structure will be completed by the end of 2024.	December, 2024

2.7 Establish a procedure to periodically review management responsibilities and accountabilities for safety issues	The AE, RE, RM, SMS Manager, and SMS Steering Committee will review management responsibilities and accountabilities annually. This annual review will be part of the recurring SMS evaluation and audit process and will include SMS Manual revisions and updates to reflect changes.	The updated SMS Manual will include information within the Assurance Section that describes an annual review to assess management responsibilities and accountabilities.	June, 2025
2.8 Establish Safety Objectives	Safety Objectives were developed as part of the draft SMS Manual developed in 2018. These will be updated to reflect regulatory requirements (Part 139 Subpart E), management requirements, PDX safety initiatives, and the new SMS Safety Policy Statement.	As part of the Policy Section in the draft 2018 SMS Manual, objectives, goals, metrics, and data sources will be updated. The VEOCI SMS Module will be available during the SMS Manual development period to verify data availability.	June, 2025

2.9 Establish a process or procedure to maintain and periodically review the Safety Objectives	SMS software will be used to monitor trends and track performance against safety objectives. Safety objectives will be reviewed periodically and revised based on SMS program performance and performance indicators. The Port intends to perform an annual review of the SMS as part of the Assurance Component and to identify opportunities for continuous improvement. The status of and changes to safety objectives will be included in the annual review.	The updated SMS Manual will include information within the Assurance Section that describes an annual review to review the safety objectives, metrics, and data.	June, 2025
2.10 Define methods, processes, and organizational structure necessary to meet Safety Objectives	The Port remains committed to providing a safe operating environment for employees, tenants, and customers. The SMS Manager will lead the effort to ensure all goals and objectives are actionable and are being monitored on frequent basis to ensure the objectives receive visibility and are being achieved. If objectives are not achieved, the SMS Manager will escalate initially to the RM to reassess and consider additional actions to ensure the objectives are being met.	The updated SMS Manual will include processes and procedures describing how objectives will be managed, monitored, and escalated.	June, 2025

3. Safety Risk Management			
3.1 Establish a system for identifying safety hazards	The Port will develop a process to identify safety hazards including the following: • Developing a customized hazard reporting form to be hosted on the PDX website and collecting data using VEOCI software. • Identify a phone number for SMS information. • Establishing a unique SMS email address. • Developing collateral material such as informational badge cards, signs, and posters and placards to support reporting, and leveraging the VEOCI software for hazard management and prioritization. The Port will employ the VEOCI software to analyze Part 139 safety hazards, incidents and accidents, trends, Operations logs, and operational changes to identify safety concerns and hazards. The Port will also develop a training and orientation plan to educate staff and tenants on reporting options, and ways to collect safety information through collaborative meetings with staff and tenants.	The updated SMS Manual will include all safety hazard reporting methods including the processes for receipt, response, tracking, and reporting.	June, 2025

3.2 Establish a systematic process to analyze hazards and their associated risks to an acceptable level	The Port will identify SRM and SRA “triggers” which will initiate the hazard reporting and analysis process. Specific triggers will include: ▪ Part 139 discrepancies that are persistent. ▪ Operational changes ▪ Incidents or accidents ▪ Hazard report submissions ▪ Construction projects and associated activities ▪ Issues identified in SMS data analyses, trends, or leading/lagging indicators	The updated SMS Manual will include the PDX risk matrix and the five-step process for analyzing hazards including how VEOCI will be leveraged to manage and report on hazard status.	June, 2025
3.3 Establish a system for regular assessment to ensure mitigations are effective	Mitigations will be tracked in VEOCI and assigned to staff to assess and monitor if the mitigations are effective. The Port will develop various reports and dashboards to monitor all hazards for recurrence.	The updated SMS Manual will include the how staff assignments and mitigation monitoring will be managed including the use of VEOCI for reporting.	June, 2025
3.4 Establish processes and procedures to document SRM efforts and retain those documents	VEOCI will serve as the central record keeping solution for the SMS.	The updated SMS Manual will include a new section entitled Record Keeping that will include all record keeping requirements and sources for reporting.	June, 2025

4. Safety Assurance			
4.1 Establish processes or procedures to monitor safety performance and safety objectives identified through safety policy	VEOCI will support the monitoring and tracking of identified objectives that have been defined as part of the Safety Policy Statement effort. Reports and dashboards will be developed including triggers to monitor each objective and performance indicator. Goals and objectives will be reviewed annually or as needed to reflect operational changes.	The updated SMS Manual will include processes and procedures for the Port to oversee objectives including use of VEOCI for trending and reporting.	June, 2025
4.2 Establish a hazard reporting system that provides a means for reporter confidentiality	The Port plans to develop an SMS hazard reporting system that aligns with federal, state, and Port confidentiality restrictions. The Port is reviewing three options available to staff and tenants for confidential reporting: 1. Anonymous reporting in which no personal information is collected. 2. Badge numbers in which only PDX security staff can identify the reporter (through SSI) 3. Full contact information in which name, email, and phone will be included. The Port will design the reporting form to align with the three options and include detailed information in the SMS orientation and training courses for staff and tenant awareness.	The hazard reporting system will be completed as part of the VEOCI SMS module implementation and will align with the updated SMS Manual prior to staff training.	June, 2025

4.3 Maintain the hazard reporting system and establish a process for reviewing and analyzing reported hazards	The hazard reporting system will be developed using the SMS Module within the VEOCI platform. VEOCI will enable the SMS team to perform the SRM five steps for hazard review including risk ranking and work order and task assignments for mitigations. The SMS/SRM module includes the ability to customize the risk matrix.	See 4.2. The updated SMS Manual will include the process for reviewing and analyzing reports including the use of VEOCI for analysis, tracking, and reporting.	June, 2025
4.4 Establish a process or procedure for reporting safety information and data on a regular basis to the Accountable Executive	Reporting safety to management, tenants, and staff will be part of the PDX overall safety communications plan and dashboards using an SMS software platform. Briefings and other communication as needed will be distributed and logged in the software. PDX plans to develop a Safety Communications Log to track communications. (See 5.7).	The updated SMS Manual will include a section describing a safety communication plan including reporting to management.	June, 2025

5. Safety Promotion			
5.1 Develop training on airport's SMS requirements specific to the audience's roles and responsibilities	The Port is in the process of evaluating existing training programs and content to identify the best options for delivering SMS-specific training and orientation. Existing training including SIDA and Airside Driver training is deployed using the Skillburst Interactive Platform. The Port will develop SMS training which will be deployed using this platform. SMS training will be completed by all badge holders with access to the airside. Additional/specialized training will be provided to staff with additional responsibilities for airside safety. Tenants will receive orientation through the existing SIDA badge training process in addition to safety briefings and committee meetings.	See 2.6. The Port will develop a staff training plan including SMS courses and orientation in time for rollout to SMS support staff in early 2026 and tenants in mid-2026.	January – March 2025

5.2 Provide initial training to personnel with roles and responsibilities in airport's SMS	The Port plans to hire an SMS Manager in mid-2024. The SMS Manager will develop a staff training plan and either deliver training in-house or engage a third party to provide the training. The training will be delivered to Port staff that will support the SMS and to Management to ensure the entire Port team has relevant skills and knowledge to manage and participate in SMS-related operations.	See 5.1. The Port will formally provide training to SMS support staff depending on roles and responsibilities in early 2025; however, the SMS development team will continuously acquire SMS knowledge beginning in 2023.	Staff: March, 2025 Tenants: July, 2025
5.3 Develop processes or procedures to record training	The Port intends to use existing training delivery and tracking systems to deliver and record training for both staff and tenants. The Port currently uses the Skillburst Interactive Platform for SIDA and driver training. This will be expanded to include SMS Orientation training for all badge holders with airside access. SMS training will be tracked, and records will be maintained using this system.	See 2.6 and 5.1. Training tracking will be defined as the staff training matrix and related courses are identified and documented.	March, 2025

5.4 Develop informational material about hazard awareness reporting	The Port plans to incorporate industry best practices to develop informational materials including the following: ▪ Signs, posters, placards, and informational badge cards with URLs ▪ SMS briefings prior to and during rollout ▪ Training courses and orientation sessions through the badging process ▪ Staff and tenant outreach through existing and new committees ▪ PDX website for resources and hosting the hazard reporting form	The Port will develop materials before the SMS support staff training and rollout scheduled for January 15, 2026. Ideally all materials will be completed, printed, and tested by no later than March 15, 2026.	March, 2026
5.5 Develop methods for communicating important safety information	The Port plans to develop a detailed Communication Plan that will outline information to be distributed to stakeholders and audiences. The Communication Plan will serve as the foundation for the safety communication record keeping effort (e.g., the Safety Communications Log). The Port is currently evaluating SMS software options for implementation. If software is unavailable, Port staff are proficient in creating forms in VEOCI and can assist in developing a log for tracking and trending.	See 4.4. The updated SMS Manual will include a safety communication plan in the Promotion section.	June, 2025

5.6 Establish a procedure to periodically review and update communication methods	The Port plans to monitor a Safety Communications Log (Log - see 5.7) and results from communication types to ensure the most effective method is employed to deliver safety messages. The Port will monitor the Log for increases in hazard reporting or reduction in safety events and will assess the communication effectiveness in the annual review.	See 2.7. The updated SMS Manual will include information within the Assurance Section that describes an annual review to assess communication effectiveness.	June, 2025
5.7 Develop processes or procedures to record safety communications	The Port plans to create a comprehensive SMS and Safety Communication Plan that includes audience, communication type, format, delivery method, and frequency. The Port will track safety communications status in a Safety Communications Log to monitor effectiveness and adjust methods or frequency. The Log will support the FAA's 12-month safety communications record keeping requirement.	The updated SMS Manual will include a new section entitled Record Keeping that will include all record keeping requirements and sources for reporting.	June, 2025